

A KoMMERCE

This section provides a detailed description of KoMMERCE, which is used to construct the MMID benchmark. KoMMERCE is a multimodal dialogue dataset built upon the K-Fashion image dataset¹ released by AIHub, and it is structured such that fashion images are naturally embedded within the dialogue flow alongside textual utterances. The dataset is designed to enable language models to integratively process textual and visual information and to engage in multi-turn dialogues that closely resemble real-world interactions.

The fashion domain represents a prototypical multimodal application area in which visual attributes and linguistic descriptions are tightly intertwined. Accurately understanding user requirements and presenting appropriate images therefore requires a joint understanding of both images and text. Accordingly, KoMMERCE structures interactions around specific fashion-related topics, where users perform actions such as querying fashion information, requesting style recommendations, or asking for images, and the assistant responds by providing corresponding explanations or fashion images.

A.1 Dialogue Structure and Annotation

To reflect realistic conversational environments, KoMMERCE features short and concise utterances per turn, and incorporates dialogue structures in which user intent or requirements are progressively expressed over multiple turns, as well as cases where consecutive short utterances appear within a single turn. These characteristics capture dialogue patterns that frequently occur in real-world user–assistant interactions.

KoMMERCE provides the following dialogue annotations: **(1) Intent and sentiment labeling:** For each user and assistant utterance, annotations are provided for both the utterance’s intent and sentiment, enabling analysis of the dialogue’s communicative purpose and emotional flow.

(2) Fashion attribute state labeling: Fashion attributes requested by users for clothing items (e.g., color, style, material) are annotated as visual attribute states throughout the dialogue. When user requirements change, the corresponding visual attribute states are reset or updated, explicitly capturing condition changes driven by the dialogue context.

(3) Cross-modal coreference annotation: When a textual utterance refers to a specific fashion image or object in the dialogue, the cross-modal coreference instance between text and image is annotated. This annotation explicitly identifies which image or object each utterance refers to, supporting multimodal reference resolution and grounding.

A.2 Dataset Statistics

Table 1 summarizes the overall statistics of KoMMERCE. The dataset consists of 15,293 dialogues, among which 12,687 contain visual attribute states and 12,670 contain cross-modal coreference instances. The dataset includes 30,792 sessions, reflecting cases in which a single dialogue spans multiple sessions. Each dialogue contains an average of 35 utterances, forming relatively long multi-turn dialogue structures. Across all dialogues, 40,461 fashion images

Table 1: Dataset statistics of KoMMERCE.

Statistic	Count
Total number of dialogues	15,293
Dialogues containing visual attribute states	12,687
Dialogues containing cross-modal coreference instances	12,670
Sessions (temporal spans of continuous dialogue)	30,792
Average number of utterances per dialogue	35
Fashion images appearing in dialogues	40,461
Unique fashion images included in dialogues	29,829
Requested visual attribute instances	58,156
Unique requested visual attributes	2,634
Average number of coreference instances per dialogue	19

appear, among which 29,829 are unique images. In addition, 58,156 requested visual attribute instances are annotated throughout the dialogues, covering 2,634 unique requested visual attributes. The average number of coreference instances per dialogue is 19.

B MMID Benchmark Construction

This section presents an overview of the MMID construction methodology and the design motivations underlying each task.

All tasks in MMID are constructed following Algorithm 1, with 200 data samples allocated for each utterance-range category. For the Yes/No setting, the numbers of samples with Yes and No as the correct answer are balanced to prevent class imbalance. Except for Tasks 1, 2, and 8, all tasks are formulated in two evaluation formats: 4-class multiple-choice questions (MCQs) and Yes/No questions, enabling analysis of differences in model inference behavior across evaluation formats. The 4-class MCQ format requires selecting the correct or incorrect option corresponding to the referent from four candidate options. The Yes/No format is designed to evaluate image–text consistency based on dialogue context by reusing one of the MCQ options and asking the model to judge the truthfulness of a given statement, attribute, or image with respect to the dialogue. To ensure consistent performance measurement across models, we constrain model outputs to select one option from A, B, C, D for 4-class MCQs and one option from YES, NO for Yes/No questions.

The procedure for constructing each task in MMID is outlined in Algorithm 3. For Tasks 3, 4, and 8, instances are built using only a subset of the full dialogue or by segmenting the dialogue into multiple paragraphs rather than providing the entire conversation. In these cases, dialogues are split based on image appearance points or dialogue topic transitions. Algorithm 4 then illustrates how task-specific instances are constructed. Using the dialogue segments and ground-truth visual attribute and images determined by this algorithm, options are generated in Algorithm 5, resulting in the final dataset instances.

Algorithm 4 further illustrates how task-specific instances are constructed. Using the dialogue segments, ground-truth annotations, and other task-specific fields defined by this algorithm, Algorithm 5 constructs candidate options and produces the final dataset instances.

¹<https://www.aihub.or.kr/aihubdata/data/view.do?currMenu=&topMenu=&aihubDataSe=data&dataSetSn=51>

Algorithm 1 Dataset Sampling with Length Buckets and Balanced Option-Type Sampling

Require: Dataset $\mathcal{D}$, task id T , target per bucket $n=200$, bucket width $w=5$, minimum bucket size $m=200$

Ensure: Sampled subset $\mathcal{S}$

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1: if USESDIALOGUE( $T$ ) = False then
2:   return RANDOMSAMPLING( $D, n \times w$ )
3: end if
    $\triangleright$  Step 0: Choose bucketing key
4:  $\mathcal{B} \leftarrow \emptyset$ 
5: for all  $x \in \mathcal{D}$  do
6:   if  $T = 8$  then
7:      $l \leftarrow \text{NUMSEGMENTS}(x) \times w$ 
8:   else
9:      $l \leftarrow \text{DIALOGUELENGTH}(x)$ 
10:  end if
11:   $b_{id} \leftarrow \lfloor l/w \rfloor$   $\triangleright b_{id}=1 \Rightarrow [1, 5], b_{id}=2 \Rightarrow [6, 10], \dots$ 
12:   $\mathcal{B}[b_{id}].\text{APPEND}(x)$ 
13: end for
    $\triangleright$  Step 1: Build buckets by 5-length intervals
14:  $b_{\max} \leftarrow \max(\text{KEYS}(\mathcal{B}))$ 
15: while  $b_{\max} > 1$  and  $|\mathcal{B}[b_{\max}]| < m$  do
16:    $\mathcal{B}[b_{\max} - 1] \leftarrow \mathcal{B}[b_{\max} - 1] \cup \mathcal{B}[b_{\max}]$ 
17:    $\text{DELETE } \mathcal{B}[b_{\max}]$ 
18:    $b_{\max} \leftarrow b_{\max} - 1$ 
19: end while
    $\triangleright$  Step 2: Merge tail buckets if the last bucket is small
20:  $C \leftarrow \text{SELECTTAILCONSECUTIVEIDS}(\mathcal{B}, T, m)$ 
21: if  $C = \emptyset$  then  $\triangleright$  global Stratified sampling by option type
22:   return BALANCEDSAMPLING( $\mathcal{D}, n, T$ )
23: end if
    $\triangleright$  Step 3: Select consecutive buckets from the tail
24:  $\mathcal{S} \leftarrow \emptyset$ 
25: for all  $b \in C$  do
26:    $\mathcal{S}_b \leftarrow \text{BALANCEDSAMPLING}(\mathcal{B}[b], n, T)$   $\triangleright$  Details in Alg.2
27:    $\mathcal{S} \leftarrow \mathcal{S} \cup \mathcal{S}_b$ 
28: end for
29: return  $\mathcal{S}$ 

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B.1 Perception Ability

Task 1: Context-aware Target Item Grounding requires the model to take a dialogue containing images introduced over multiple turns as input and, at the final stage of the dialogue, identify the clothing item that corresponds to the user’s requested conditions among the recommended images, and output the spatial location of that item as image coordinates. Specifically, this task evaluates whether the model can accurately recognize the clothing item requested by the user from the dialogue context and correctly map it to its spatial location within the image.

Question & Option Design. Unlike other tasks that are formulated as 4-class MCQs or Yes/No questions, Task 1 is designed to require direct generation of grounding locations within an image. Each image in KoMMERCE is annotated with bounding box coordinates $[x, y, w, h]$ corresponding to the clothing item requested by the user,

Algorithm 2 BALANCEDSAMPLING($\mathcal{X}, n, T$) (Option-Type-Stratified Sampling)

Require: Candidate instance set $\mathcal{X} = \{x_1, \dots, x_M\}$, target size n , task id T

Ensure: Sampled subset $\mathcal{Y}$

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1: if  $|\mathcal{X}| \leq n$  then
2:   return  $\mathcal{X}$ 
3: end if
    $\triangleright$  Group instances by task-dependent option type
4:  $\mathcal{G} \leftarrow \text{GROUPBYOPTIONTYPE}(\mathcal{X}, T)$ 
5:  $\mathcal{T} \leftarrow \text{KEYS}(\mathcal{G})$ 
6:  $K \leftarrow |\mathcal{T}|$ 
7:  $\mathcal{Y} \leftarrow \emptyset$ 
    $\triangleright$  Allocate near-uniform quota per option type
8:  $q \leftarrow \lfloor n/K \rfloor$ 
9: for all  $t \in \mathcal{T}$  do
10:   $k_t \leftarrow \min(q, |\mathcal{G}[t]|)$ 
11:   $\mathcal{Y} \leftarrow \mathcal{Y} \cup \text{RANDOMSAMPLE}(\mathcal{G}[t], k_t)$ 
12: end for
    $\triangleright$  Fill remaining slots from the leftover pool
13:  $\mathcal{R} \leftarrow \mathcal{X} \setminus \mathcal{Y}$ 
14:  $\mathcal{Y} \leftarrow \mathcal{Y} \cup \text{RANDOMSAMPLE}(\mathcal{R}, n - |\mathcal{Y}|)$ 
15: return  $\mathcal{Y}$ 

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Algorithm 3 Task Instance Construction

Require: Dialogue D , Task index T

Ensure: Constructed instance $\mathcal{I}$

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    $\triangleright$  Step 1: Dialogue preprocessing
1: if  $T \in \{3, 4, 8\}$  then
2:    $D' \leftarrow \text{SPLITDIALOGUE}(D)$ 
3: else
4:    $D' \leftarrow D$ 
5: end if
    $\triangleright$  Step 2: Initialize instance fields
6:  $\mathcal{I} \leftarrow \{\}$ 
7:  $\mathcal{I}.task\_id \leftarrow T$ 
8:  $\mathcal{I}.dialogue \leftarrow D'$ 
9:  $\mathcal{I}.options \leftarrow \emptyset$ 
10:  $\mathcal{I}.gold \leftarrow \emptyset$ 
    $\triangleright$  Step 3: Populate task-specific fields (Alg. 4)
11:  $\mathcal{I} \leftarrow \text{BUILDTASKFIELDS}(\mathcal{I})$ 
    $\triangleright$  Step 4: Finalize instance with options and answer mapping (Alg. 5)
12:  $\mathcal{I} \leftarrow \text{FINALIZEINSTANCE}(\mathcal{I})$ 
13: return  $\mathcal{I}$ 

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which are used as the ground truth. The model is provided with a dialogue that includes images, and the question is formulated to prompt the model to predict and output the corresponding coordinates for the clothing item the user intends to be recommended at the end of the dialogue.

Task 2: Text-to-Image Attribute Matching takes textual descriptions of visual attributes as input and requires the model to select the clothing image that best matches those attributes from a set of candidate images. This task evaluates whether the model can align

Algorithm 4 BUILDTASKFIELDS(I)**Require:** Instance I with $I.task_id = T$ and $I.dialogue = D$ **Ensure:** Updated instance I

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1:  $D \leftarrow I.dialogue$ 
2: if  $T = 1$  then
3:    $I \leftarrow \text{LASTIMAGE}(D)$ 
4:    $I.gold \leftarrow \text{EXTRACTKEYITEMBBOX}(I)$ 
5: else if  $T = 2$  then
6:    $I \leftarrow \text{RANDOMSELECTIMAGE}(D)$ 
7:    $I.tag\_text \leftarrow \text{EXTRACTTAGS}(I)$ 
8:    $I.gold \leftarrow I$ 
9: else if  $T = 3$  then
10:   $D \leftarrow \text{FIRSTDIALOGUE}(D)$ 
11:   $I \leftarrow \text{LASTIMAGE}(D)$ 
12:   $D \leftarrow \text{ERASELASTIMAGE}(D)$ 
13:   $I.gold \leftarrow I$ 
14: else if  $T = 4$  then
15:   $D \leftarrow \text{LASTDIALOGUE}(D)$ 
16:   $I \leftarrow \text{LASTIMAGE}(D)$ 
17:   $D \leftarrow \text{ERASELASTIMAGE}(D)$ 
18:   $I.gold \leftarrow I$ 
19: else if  $T = 5$  then
20:   $S \leftarrow \text{RANDOMSELECTSITUATION}(D)$ 
21:   $I \leftarrow \text{LASTIMAGE}(D, S)$ 
22:   $I.tag\_text \leftarrow \text{EXTRACTTAGS}(I)$ 
23:   $u_{\text{virtual}}, gold\_tags \leftarrow \text{GENERATEUSERUTTERANCE}(S, I.tag\_text)$ 
24:   $D \leftarrow \text{APPEND}(D, u_{\text{virtual}})$ 
25:   $I.gold\_tags \leftarrow gold\_tags$ 
26:   $I.gold \leftarrow \text{SEARCHIMAGEBYTAGS}(gold\_tags)$ 
27: else if  $T = 6$  then
28:   $(\mathcal{X}, preferences) \leftarrow \text{GETIMAGESANDPREF}(D)$ 
29:   $I.images \leftarrow \mathcal{X}$ 
30:   $I.preferences \leftarrow preferences$ 
31: else if  $T = 7$  then
32:   $(indicator, I) \leftarrow \text{GETINDICATORANDTARGETIMAGE}(D)$ 
33:   $I.indicator \leftarrow indicator$ 
34:   $I.gold \leftarrow I$ 
35: else if  $T = 8$  then
36:   $I_{\text{seq}} \leftarrow \text{CORRECTSEQIMAGES}(D)$ 
37:   $(D, gold\_seq) \leftarrow \text{RANDOMSEQUENCE}(D)$ 
38:   $I.img\_seq \leftarrow I_{\text{seq}}$ 
39:   $I.gold \leftarrow gold\_seq$ 
40: end if
41:  $I.dialogue \leftarrow D$ 
42: return  $I$ 

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attributes expressed in text with corresponding visual attributes in images.

Question & Option Design. Data samples for Task 2 are constructed based on visual attribute annotations validated in KoMERCe. For each dialogue, the final set of visual attributes requested by the user is extracted, and the corresponding image options, together with a fixed question template, are used as a single input. The number of input attributes ranges from 2 to 9, with an average of approximately 4 attributes per sample. All questions

Algorithm 5 FINALIZEINSTANCE(I)**Require:** Instance I with $I.task_id$ and task-specific fields**Ensure:** Updated instance I with options and answer mapping

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1:  $T \leftarrow I.task\_id$ 
    $\triangleright$  Option construction and answer mapping depend on task type
2: if  $T = 1$  then  $\triangleright$  Bounding-box task (no multiple-choice options)
3:    $I.answer \leftarrow I.gold\_bbox$ 
4: else if  $T \in \{2, 3, 5, 7\}$  then  $\triangleright$  Image-option tasks
5:    $O, O_{\text{types}} \leftarrow \text{SAMPLEIMAGEOPTIONS}(I)$   $\triangleright$  includes gold image
6:    $O, O_{\text{types}} \leftarrow \text{SHUFFLE}(O, O_{\text{types}})$ 
7:    $I.options \leftarrow O$ 
8:    $I.option\_types \leftarrow O_{\text{types}}$ 
9:    $I.answer \leftarrow \text{INDEXOF}(O, I.gold)$ 
10: else if  $T \in \{4, 6\}$  then  $\triangleright$  Text-option tasks
11:    $O, O_{\text{types}} \leftarrow \text{BUILDTTEXTOPTIONS}(I)$   $\triangleright$  includes correct statement
12:    $O, O_{\text{types}} \leftarrow \text{SHUFFLE}(O, O_{\text{types}})$ 
13:    $I.options \leftarrow O$ 
14:    $I.option\_types \leftarrow O_{\text{types}}$ 
15:    $I.answer \leftarrow \text{INDEXOF}(O, I.gold)$ 
16: else if  $T = 8$  then  $\triangleright$  Sequence ordering task
17:    $O \leftarrow \text{BUILDSEQUENCEOPTIONS}(I)$   $\triangleright$  candidate permutations
18:    $I.options \leftarrow O$ 
19:    $I.answer \leftarrow I.gold\_seq$ 
20: end if
21: return  $I$ 

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use a fixed response format to restrict free-form generation and ensure consistent performance measurement.

Task 2 is formulated in both 4-class MCQ and Yes/No question formats. In the 4-class MCQ setting, one correct image and three incorrect images are provided as options. In the Yes/No setting, the model judges whether a single image is consistent with the given visual attributes. Incorrect images are generated by keeping the clothing category and mid-level classification identical to those of the correct image, while modifying only one fine-grained (sub-category-level) attribute. For example, if the correct image satisfies the attribute “Top: Length: Long”, the incorrect image is constructed by preserving the higher-level categories, Top and Length, and changing only the fine-grained attribute to “Top: Length: Crop”. By constraining each incorrect image to contain only one attribute change, the task ensures that incorrect selections directly correspond to failures in recognizing specific visual attributes.

Task 3: Incremental Requirement-based Image Selection takes as input a multi-turn, text-only user–assistant dialogue and requires the model to select a clothing image that satisfies the user’s request conditions revealed throughout the dialogue. Unlike Task 2, which directly provides explicit attribute information as input, Task 3 requires the model to progressively recognize and integrate user requirements that are distributed across the entire multi-turn dialogue flow.

Question & Option Design. In KoMMERCE dialogues, the assistant asks attribute-specific questions—such as garment length or color—to infer user preferences, and the user provides desired clothing conditions either across multiple turns or through consecutive short utterances within a single turn. In Task 3, only the textual utterances up to the point before the assistant presents any image are used as input, and the images subsequently provided by the assistant are used to construct the answer options.

Based on the dialogue flow, Task 3 is divided into two sub-types. Type 1 corresponds to cases in which images are presented for the first time after the user has provided requirements across multiple turns. Type 2 corresponds to cases in which the user provides additional conditions even after images have been presented, resulting in longer dialogue contexts with accumulated user requirements. Task 3 is formulated in both 4-class MCQ and Yes/No question formats, and the basic strategy for constructing incorrect images follows that of Task 2.

However, in the Type 1 setting, the amount of attribute information available at the image selection point is relatively limited. As a result, simple visual attribute modifications are often insufficient to clearly distinguish correct and incorrect options, and multiple options may appear to simultaneously satisfy the user’s conditions. This increases the relative difficulty of comparing option images compared to Type 2. To address this limitation, the 4-class MCQ setting for Type 1 is designed such that the three incorrect options each reflect a distinct error type. Specifically, one incorrect option corresponds to a visual attribute modification–based error, while the other two correspond to an under-editing error, where some required attributes mentioned in the dialogue are not reflected, and an over-editing error, where attributes not requested in the dialogue and not present in the correct image are added. These error types enable analysis of both failures in recognizing visual attributes and failures in accurately integrating user requirements.

Task 4: Fine-grained Visual Attribute Verification is based on dialogues that include a single image and requires the model to determine whether a textual description correctly matches the fine-grained visual attributes of an image presented in the dialogue. While Tasks 2 and 3 evaluate the process of selecting images based on textually provided attribute information, Task 4 requires the inverse perceptual process of extracting visual attributes from images and comparing them against attribute-level linguistic descriptions. For this reason, Tasks 2–4 are designed to complementarily evaluate the model’s perception ability.

Question & Option Design. We extract KoMMERCE data samples in which the assistant provides only a single image and the dialogue ends with the user expressing preference for that image. For each image, one of the labeled fine-grained attributes is selected, and a sentence corresponding to that attribute is generated. Sentences aligned with the image directly reflect the labeled attributes. For example, if an image is labeled with the attributes “Top: Category: Shirt” and “Top: Length: Long”, a sentence such as “The user prefers a long-length shirt.” is generated.

In contrast, non-aligned sentences are generated by modifying only the fine-grained attribute while preserving the same category and mid-level classification. For instance, “Top: Length: Long” is changed to “Top: Length: Crop” to produce a sentence such as “The

user prefers a cropped-length shirt.” Task 4 is formulated in both 4-class MCQ and Yes/No question formats.

B.2 Reasoning Ability

Task 5: Selective Preference Recall and Editing takes as input a dialogue that contains two or more distinct scenarios and requires the model to select the most appropriate image when the requirement conditions of a specific scenario are modified based on the user’s final utterance. While Task 3 evaluates the model’s ability to aggregate user requirements across dialogue turns, Task 5 extends this evaluation by assessing whether the model can identify the scenario indicated by the final user utterance and recall and selectively update the requirement conditions associated with that scenario.

Question & Option Design. We first extract dialogues from KoMMERCE that contain two or more scenarios and identify the start and end boundaries of each scenario within a dialogue. One scenario is then randomly selected, and a single visual attribute is extracted from the user requirements mentioned in that scenario. Next, a synthetic user utterance that modifies the extracted attribute is generated and appended to the end of the dialogue. The correct option is constructed as an image that reflects the updated attribute in accordance with the final user requirements; specifically, one image is randomly selected from the set of images that satisfy all updated requirements.

In addition, one incorrect option is constructed for each error type—Session Confusion, Under-Editing, and Over-Editing—enabling systematic analysis of whether the model makes errors at the stage of scenario identification, requirement recall, or attribute updating.

- *Session Confusion*: An image selected based on user requirements from a different scenario in the dialogue, rather than the scenario indicated by the final utterance.
- *Under-Editing*: An image in which the correct scenario is identified, but the attribute modification requested in the final utterance is partially or not reflected at all.
- *Over-Editing*: An image in which attributes other than those explicitly requested in the final utterance are additionally modified.

Task 6: Implicit Preference Reasoning takes as input a dialogue containing multiple images and requires the model to judge the truthfulness of a sentence by jointly considering fine-grained visual attributes of each image and the user’s preferences toward those images. While Task 4 evaluates visual attribute perception for a single image, Task 6 extends this setting to assess the model’s ability to distinguish attributes across multiple images and integratively reason about user preferences based on dialogue context.

Question & Option Design. Task 6 is formulated in both 4-class MCQ and Yes/No question formats. The 4-class MCQ includes two question types: Most Appropriate and Least Appropriate. The Most Appropriate format requires selecting the single most correct option among four candidates, consisting of one correct option and three incorrect options constructed based on different error types. This design enables simultaneous analysis of multiple error types within a single query instance. Task 6 is further divided into two sub-types

according to difficulty and evaluation objectives, focusing on image-specific attribute recall and preference recognition in multi-image settings.

Type 1 evaluates whether the model can accurately recall fine-grained attributes of multiple images that the user explicitly mentions as preferred in the dialogue. Options are constructed as sentences corresponding to attributes extracted from each preferred image, and sentences aligned with both the image and its attributes are treated as aligned (correct) sentences. For example, if a preferred image is annotated with the attributes “Top: Category: Shirt” and “Top: Color: Red”, the aligned sentence is “The user prefers a red shirt.” To evaluate attribute perception errors, hallucinated attribute generation, and preference misinterpretation, the following non-aligned sentences are generated:

- *Positive Mid Attribute Editing*: A sentence constructed by injecting an attribute that does not exist in the image (e.g., “Top: Print: Floral”), such as “The user prefers a floral-print shirt.”
- *Positive Intra Attribute Editing*: A sentence constructed by modifying an attribute of a preferred image (e.g., “Top: Color: Red” → “Top: Color: Orange”), such as “The user prefers an orange shirt.”
- *User Sentiment Change*: A sentence in which image attributes are preserved but the user preference is reversed (e.g., “The user does not prefer a red shirt.”)

Type 2 evaluates whether the model can clearly distinguish between preferred and non-preferred images within the same scenario, requiring a higher level of reasoning than Type 1. While Type 1 focuses on independently recalling attribute information for multiple preferred images, Type 2 requires jointly comparing attributes and user preferences across both preferred and non-preferred images to judge sentence truthfulness. To this end, dialogues containing one preferred image and multiple non-preferred images are used to construct options. Options are generated based on attribute differences between preferred and non-preferred images, with shared attributes (excluding category-level attributes) removed to ensure that each sentence corresponds to a specific image. For example, if the preferred image has attributes “Style: Modern”, “Top: Color: Red”, “Top: Length: Crop”, and “Top: Category: Shirt”, and the non-preferred image has “Style: Modern”, “Top: Color: Green”, “Top: Length: Long”, and “Top: Category: Shirt”, the shared attribute “Style: Modern” is removed. The aligned options may then be constructed as follows: “The user prefers a red shirt.”, “The user does not prefer a green shirt.”, “The user prefers a cropped-length shirt.”, and “The user does not prefer a long-length shirt.” As in Type 1, we apply the *User Sentiment Change* error type by generating sentences with reversed user preferences to evaluate preference recognition ability. In addition, by cross-swapping attributes between preferred and non-preferred images to generate non-aligned sentences, we assess whether the model can discriminate images based on user preferences.

- *Positive2Negative Attribute Transfer*: A sentence in which attributes of a preferred image are replaced with those of a non-preferred image. (e.g., “Top: Color: Red” → “Top: Color: Green”: “The user prefers a green shirt.”)

- *Negative2Positive Attribute Transfer*: A sentence in which attributes of a non-preferred image are replaced with those of a preferred image (e.g., “The user does not prefer a cropped-length shirt.”)

Task 7: Visual Reference Recall takes as input a dialogue containing multiple images and requires the model to identify the reference image indicated by deictic expressions appearing in the user’s utterances. The MLLM must identify the referent of the deictic expression based on the dialogue context and recall previously mentioned image information to select the appropriate image.

Question & Option Design. We first extract dialogues from KoM-MERCE that contain multiple images and then sample those that include deictic expressions grounded in prior dialogue context. To prevent the correct answer from being inferred solely from textual information, we modify cases where clothing categories and images exhibit a one-to-one correspondence by removing explicit category information from the deictic expression and instead incorporating visual attributes of the image. For example, expressions such as “this shirt” are transformed into “this blue item” to reflect image-specific visual attributes.

The correct option corresponds to the image referred to by the deictic expression, while incorrect options are constructed to reflect different error types depending on the number of images in the dialogue. When the dialogue contains two images, we include one incorrect option of type *Different Image in Dialog* and two incorrect options of type *Different Attribute*. When the dialogue contains three images, we include two *Different Image in Dialog* incorrect options and one *Different Attribute* incorrect option. This design enables systematic analysis of whether the model fails at the stage of deictic interpretation and reference resolution, or at the stage of recalling image-specific attributes.

- *Different Image in Dialog*: An image appearing in the dialogue that is not the referent of the deictic expression.
- *Different Attribute*: An image in which the correct referent is identified, but some visual attributes are incorrectly recalled or modified.

Task 8: Vision–Language Temporal Reconstruction requires reconstructing randomly shuffled dialogue segments into their correct temporal order by leveraging accompanying image(s) as visual cues. This task evaluates whether the model can integrate visual information with dialogue context to infer the temporal progression of the conversation.

Question & Option Design. We extract dialogues from KoM-MERCE that contain at least three images. Each dialogue is segmented into dialogue paragraphs according to image presentation boundaries, and each paragraph is mapped to labels such as (a), (b), and so on. The image information is then removed, and the paragraph order is randomly shuffled to form the input. Each option is constructed as an ordered sequence of paragraph labels, with the correct option reflecting the original dialogue order. Incorrect options are generated by randomly permuting the paragraph order.

Task 8 is divided into two sub-types according to the manner in which image cues are provided. Type 1 provides all images in accordance with the dialogue flow, allowing the model to leverage alignment between images and dialogue progression to infer temporal order. Type 2 increases the difficulty by providing only the

image appearing in the final paragraph. In this setting, the model

must infer the entire dialogue context and reconstruct the paragraph order using limited visual cues, thereby requiring a higher level of temporal reasoning ability than Type 1.